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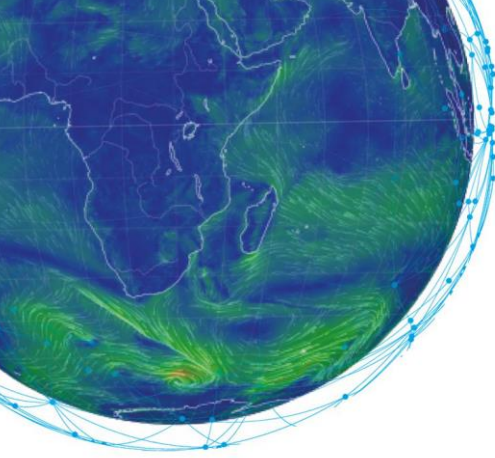
MOOC

Machine Learning in Weather & Climate

TRANSCRIPT

**WeatherBench as a benchmark
dataset**

Expert: Mariana Clare



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WeatherBench as a benchmark dataset

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Introduction

After learning a bit about the data within the WeatherBench benchmark dataset, we are now going to see how the WeatherBench dataset meets the criteria of being a good benchmark dataset. Let's return once again to Mariana Clare at the ECMWF headquarters in Bonn, Germany.

Script

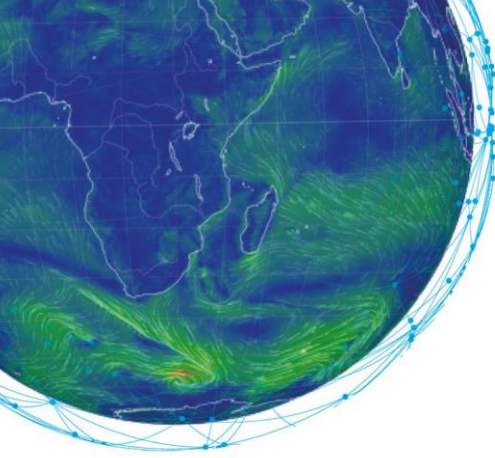
The preprocessed data is hosted on a public repository and a starter code, including code to evaluate the task set by WeatherBench and visualise results, is available in a public code repository on GitHub. Hence WeatherBench meets the criteria of being an open source, easy-to-use dataset, with code for visualisations and evaluation metrics.

WeatherBench was inspired by the evaluation done at operational weather centres like ECMWF. It sets the task of predicting atmospheric variables such as geopotential height at 500hPa and temperature at 850hPa. These variables are a good measure for how well the models capture the dynamical evolution of the atmosphere. In particular, geopotential height at constant pressure levels like 500hPa can be used to identify weather systems such as cyclones and anticyclones.

It also sets the task of predicting the surface variables of total precipitation over a 6 hour period and temperature at 2m. Considering the temperature at this height is standard practice in weather models, to reduce the influence of land warming effects on the temperature.

WeatherBench then uses two key well established metrics, namely area-weighted root mean squared error and the anomaly correlation coefficient, to evaluate model performance.

When developing new techniques, whether they be numerical models or a neural network, baselines are crucial as reference points to judge whether a new model is performing well. WeatherBench includes a persistence forecast, which assumes that future weather will be the same as current weather i.e. that the weather will “persist”; and a climatology which is based on average conditions over the last 40 years.



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Next we have baselines from physics-based dynamical models, which are used for operational weather forecasting today. ECMWF's IFS forecast (see purple line) provides the gold standard, while lower resolution dynamical models provide intermediate targets.

In addition, WeatherBench includes some simple machine learning baselines from a linear regression and simple CNN, as shown in the figure. Therefore WeatherBench has all the attributes of a good benchmark dataset.

Since the publication of WeatherBench, several studies have been published with new scores, which can be seen on the official leaderboard on GitHub. However, one noticeable fact is that despite WeatherBench being more than 2 years old at this stage, only very few submissions have been made. This highlights one of the difficulties of designing benchmarks in the weather and climate space which is that there are many different subproblems and the number of researchers interested in each subproblem is often small. There is also a significant technical hurdle associated with large data volumes.

Summary

You have now learnt how even a well-constructed benchmark dataset like WeatherBench can have problems. Despite this there has been many interesting machine learning models applied to WeatherBench, which you will learn about in the following e-learning module. In the meantime, if you have any questions or would like to discuss this subject in further detail, please use our Forum on Moodle. It will be a pleasure to hear from you.